1. What is Machine Learning?

Machine learning (ML) is a branch of artificial intelligence in which computers learn patterns from data instead of being programmed with explicit rules. A classic definition by Tom Mitchell (1997) says a program learns from experience E with respect to a task T and a performance measure P if its performance on T, measured by P, improves with E.

There are three main types of machine learning. In supervised learning, the model is trained on labelled examples, where every input comes with the correct output, for example emails labelled as spam or not spam. In unsupervised learning, the data has no labels and the model must discover structure on its own, such as grouping customers into segments with clustering. In reinforcement learning, an agent learns by interacting with an environment and receiving rewards or penalties, which is how game-playing systems such as AlphaGo were trained.

A typical ML workflow has six steps: collect data, clean and prepare it, choose a model, train the model, evaluate it on unseen data, and finally deploy and monitor it. Data preparation usually takes the most time; practitioners often say it is about 80 percent of the work.

2. Common Supervised Learning Algorithms

Linear regression predicts a continuous number, such as a house price, by fitting a straight line (or plane) that minimises the mean squared error between predictions and true values. It is simple, fast and easy to interpret, which makes it a good first baseline.

Logistic regression is used for classification, not regression, despite its name. It passes a linear combination of the inputs through the sigmoid function, which squashes any value into a probability between 0 and 1. If the probability is above a threshold, usually 0.5, the example is assigned to the positive class.

A decision tree splits the data with a series of yes/no questions on the features, choosing each split to make the resulting groups as pure as possible, measured by Gini impurity or entropy. Trees are easy to visualise but overfit easily when grown too deep.

A random forest trains many decision trees on random subsets of the rows and features and combines their votes. This technique, called bagging, reduces variance and usually gives much better accuracy than a single tree. The k-nearest neighbours (KNN) algorithm classifies a new point by the majority label of the k closest training points; it has no training phase but becomes slow on large datasets.

3. Overfitting, Underfitting and Validation

Overfitting happens when a model memorises the training data, including its noise, so it scores very well on training data but poorly on new data. Underfitting is the opposite: the model is too simple to capture the real pattern and performs badly on both training and test data.

This is described by the bias-variance trade-off. High bias means the model makes strong, wrong assumptions and underfits. High variance means the model is too sensitive to small changes in the training data and overfits. The goal is a model with low total error, which requires balancing the two.

Ways to reduce overfitting include collecting more training data, using a simpler model, regularisation (L1 / Lasso adds the absolute size of the weights to the loss, L2 / Ridge adds their squares), dropout in neural networks, and early stopping when validation error starts to rise.

To measure performance honestly, data is split into training, validation and test sets; a common split is 70/15/15. In k-fold cross-validation the data is divided into k equal folds (often k = 5 or 10); the model is trained k times, each time holding out a different fold for validation, and the k scores are averaged.

4. Evaluation Metrics for Classification

Accuracy is the share of predictions that are correct. It is misleading on imbalanced data: if only 1 percent of transactions are fraud, a model that always predicts 'not fraud' is 99 percent accurate but useless.

A confusion matrix counts four outcomes: true positives (TP), false positives (FP), true negatives (TN) and false negatives (FN). Precision = TP / (TP + FP) answers: of everything the model flagged as positive, how much really was positive? Recall = TP / (TP + FN) answers: of all real positives, how many did the model find?

Precision matters most when false alarms are expensive, for example a spam filter that must not hide important emails. Recall matters most when missing a case is dangerous, for example cancer screening. The F1 score is the harmonic mean of precision and recall, F1 = 2 x (precision x recall) / (precision + recall), and is useful when you need a single number that balances both.

The ROC curve plots the true positive rate against the false positive rate at every threshold. The area under it, called AUC, ranges from 0.5 (random guessing) to 1.0 (a perfect classifier).